Software Requirements Specification (SRS)

Hospital Management System

Item	Details
Project	Hospital Management System (HMS)
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Revision history

Version	Date	Author	Change summary	Approval
1.0	28-09-2026	HMS Project Team	Compact SRS aligned to the same baseline; corrected author/SRN mapping and condensed repeated material.	Pending review

Approvals

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Table of Contents

Section	Page(s)
1. Introduction	2
2. Overall description	3
3. External interface requirements	4
4. System features (detailed)	5-8
5. Non-functional & security requirements	9-10
6. Quality attributes & acceptance criteria	10
7. System models and use cases	11-13
8. Requirements Traceability Matrix (RTM)	14-16

1. Introduction

1.1 Purpose

This SRS defines the functional, non-functional and security requirements, external interfaces, use cases, validation rules and verification criteria for the Hospital Management System (HMS). It uses the course IEEE-style SRS structure and serves as the requirements baseline for the Test Plan and Software Architecture & Design Specification.

1.2 Scope

HMS is a standalone C++17 console application used by Hospital Staff / Administrator on one local machine. It manages patient records, doctor records and appointments; prevents double-booking of a doctor at the same date and time; validates input; and stores records in local CSV files.

- In scope: register/search/view/update patients; add/search/view/update doctors; book/view/cancel appointments; availability conflict validation; CSV persistence; input validation; safe handling of missing/malformed files.
- Out of scope: billing/payments, pharmacy, laboratory, insurance, AI diagnosis, ambulance, online consultation, web/mobile/cloud deployment, networking, database servers, microservices and multi-user concurrency.
- Optional only: Delete Patient, Reschedule Appointment, Basic Reports and Administrator Login.

1.3 Audience

Project developers/testers, course instructor and evaluators. Sections 4-5 support implementation/testing; Sections 7-8 support review, traceability and test derivation.

1.4 Definitions and conventions

Term	Meaning
Hospital Staff / Administrator	Sole actor operating the console application.
Patient / Doctor	System entities managed by staff; not direct actors.
Conflict	A second Booked appointment for the same doctor at the same date and time.
CSV	Comma-separated values text file.
FR / NFR / SR / SO	Functional / Non-functional / Security Requirement / Security Objective.
UC / BR / VR / TC / RTM	Use Case / Business Rule / Validation Rule / Test Case / Requirements Traceability Matrix.

1.5 References

HMS Project Design Report - Final Baseline v1.0 (Frozen); course SRS template; IEEE Std 830-1998; ISO/IEC/IEEE 29148:2018.

2. Overall description

2.1 Product perspective and major functions

Logical flow: User -> Console UI -> HospitalSystem -> FileManager -> patients.csv / doctors.csv / appointments.csv. The application is self-contained and has no external business-system integration.

Component	Responsibility
Console UI	Displays menus, accepts input and presents results/errors.
HospitalSystem	Coordinates patient, doctor and appointment operations; applies business rules.
FileManager	Loads/saves persistent records in CSV files.
Data files	patients.csv, doctors.csv and appointments.csv persist data across restarts.

2.2 User role and entities

Item	Role / key attributes
Hospital Staff / Administrator	Sole actor; basic computer skills; uses one machine at a time.
Patient	Patient ID, name, age, gender, contact, address, optional medical-history note.
Doctor	Doctor ID, name, specialization, department, contact, availability text.
Appointment	Appointment ID, Patient ID, Doctor ID, date, time, status (Booked/Cancelled).

2.3 Operating environment and constraints

- Windows 10 or later or Linux; GCC/MinGW-w64 compatible C++17 toolchain; local read/write data directory.
- C-01 C++ with OOP/STL. C-02 Console/menu UI only. C-03 CSV/delimited-text persistence only. C-04 Single-user, single-process; no networking. C-05 Text containing commas is rejected. C-06 Mandatory scope follows Frozen Baseline v1.0.
- A-01 One operator/instance at a time. A-02 Staff identify patients correctly. A-03 Doctor availability text is informational; enforced availability uses conflicts only. A-04 Conflict means identical doctor/date/time for a Booked appointment. A-05 Data directory permissions are provided by the OS. A-06 No system-clock rule is mandatory. A-07 Patient/Doctor IDs are staff-assigned; Appointment IDs are system-generated.

2.4 Business rules

ID	Rule	Mapped requirements
BR-01	Patient IDs shall be unique.	HMS-F-001, HMS-F-002, HMS-F-007, HMS-F-028
BR-02	Doctor IDs shall be unique.	HMS-F-008, HMS-F-009, HMS-F-014, HMS-F-028
BR-03	An appointment shall require an existing Patient ID.	HMS-F-015, HMS-F-016, HMS-F-028
BR-04	An appointment shall require an existing Doctor ID.	HMS-F-015, HMS-F-017, HMS-F-028
BR-05	A doctor shall not have two non-cancelled appointments at the same date and time.	HMS-F-015, HMS-F-019, HMS-F-020, HMS-F-021
BR-06	Cancelling an appointment shall change its status to Cancelled rather than deleting its record.	HMS-F-021, HMS-F-023, HMS-F-024, HMS-SR-006
BR-07	(Optional feature) A patient with existing appointments shall not be deleted.	HMS-O-001
BR-08	Input shall be validated before any record is modified.	HMS-F-003, HMS-F-006, HMS-F-010, HMS-F-013, HMS-F-018, HMS-F-033, HMS-F-034, HMS-SR-001
BR-09	If a required data file does not exist, the corresponding dataset shall be safely treated as empty.	HMS-F-025, HMS-F-026
BR-10	Malformed records shall be skipped safely rather than causing the application to crash.	HMS-F-027, HMS-SR-004
BR-11	(Optional feature) Rescheduling shall use the same conflict validation as booking.	HMS-O-002

3. External interface requirements

3.1 User interface

Menu	Options
Main	1 Patient Management; 2 Doctor Management; 3 Appointment Management; 4 Exit
Patient	1 Register; 2 Search; 3 View All; 4 Update; 0 Return
Doctor	1 Add; 2 Search; 3 View All; 4 Update; 0 Return
Appointment	1 Book; 2 View; 3 Cancel; 4 Check Doctor Availability; 0 Return

3.2 Hardware / software / communications

- Hardware: standard keyboard/display and local disk storage; no special peripherals.
- Software: OS file system and C++17 standard library; no third-party library, database or external service required by the mandatory baseline.
- Communications: none; the application does not use a network interface.

3.3 Persistent file layouts

File	Columns (in order)	Unique key
patients.csv	PatientID, Name, Age, Gender, Contact, Address, MedicalHistoryNote	PatientID
doctors.csv	DoctorID, Name, Specialization, Department, Contact, Availability	DoctorID
appointments.csv	AppointmentID, PatientID, DoctorID, Date, Time, Status	AppointmentID

4. System features - Table 4-1: Field validation rules

Rule	Field	Validation
VR-01	Patient ID / Doctor ID	1–10 characters; letters and digits only; compared case-insensitively and stored in upper case.
VR-02	Name (patient, doctor)	1–50 characters; letters, spaces, full stop, hyphen and apostrophe only.
VR-03	Age	Whole number from 0 to 120.
VR-04	Gender	Single letter M, F or O (case-insensitive; stored in upper case).
VR-05	Contact number	Exactly 10 digits.
VR-06	Address	1–100 characters.
VR-07	Medical history note (optional)	0–200 characters; may be left empty. Plain free-text attribute of the patient record.
VR-08	Specialization, Department	1–40 characters each.
VR-09	Availability information	1–60 characters of free text (e.g., “Mon-Fri 09:00-17:00”). Informational only (see Section 2.6).
VR-10	Appointment date	Format YYYY-MM-DD; a valid calendar date (leap years honoured); year 2000–2100.
VR-11	Appointment time	Format HH:MM, 24-hour clock, 00:00–23:59.
VR-12	Appointment ID	System-generated A + at least four digits (A0001, A0002, ...); unique.
VR-13	Appointment status	Exactly one of: Booked, Cancelled.
VR-14	Text-safety rule (all text fields)	Value shall not contain a comma, carriage-return or line-feed character, and shall not begin with =, +, - or @.

4.1 Patient Management - Functional Requirements

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-001	The system shall allow Hospital Staff to register a patient by entering Patient ID, name, age, gender, contact number, address and an optional medical history note.	Functional	High	Hospital Staff; UC-01; BR-01	Valid data creates exactly one new patient record, a confirmation is displayed, and patients.csv contains the record after the operation. Test: TC-PAT-01	HospitalSystem, Patient, FileManager
HMS-F-002	The system shall reject registration of a patient whose Patient ID already exists and shall leave all stored records unchanged.	Functional	High	Hospital Staff; UC-01; BR-01	Registering an existing ID (in any letter case) displays “Patient ID already exists”; patient count and file content are identical before and after. Test: TC-PAT-02	Depends on HMS-F-001
HMS-F-003	The system shall validate every patient field against rules VR-01 to VR-07 and VR-14 before storing it, and shall reject an invalid value with a message naming the field and the rule violated.	Functional	High	Hospital Staff; UC-01; BR-08	Boundary values (minimum, maximum, maximum + 1) for each field are accepted or rejected as defined in Table 4-1; no record is stored while any field is invalid. Test: TC-PAT-03	Table 4-1
HMS-F-004	The system shall allow Hospital Staff to search patients by Patient ID (exact match) or by name (case-insensitive partial match) and shall display all matching patients.	Functional	High	Hospital Staff; UC-02	Search “P001” returns only that patient; search “sha” returns every patient whose name contains “sha”; no match displays “No matching patient found”. Test: TC-PAT-04	—
HMS-F-005	The system shall display all stored fields of a selected patient and shall provide a list of all registered patients showing Patient ID, name, age, gender and contact number.	Functional	High	Hospital Staff; UC-02	Selected patient view shows all seven fields exactly as stored; list view shows every stored patient once; an empty dataset displays “No patients registered”. Test: TC-PAT-05	Depends on HMS-F-001
HMS-F-006	The system shall allow Hospital Staff to update the name, age, gender, contact number, address and medical history note of an existing patient, applying the same validation rules as registration.	Functional	Medium	Hospital Staff; UC-03; BR-08	A valid change is reflected on the next view and in patients.csv; an invalid value is rejected and the stored record is unchanged. Test: TC-PAT-06	Depends on HMS-F-001, HMS-F-003
HMS-F-007	The system shall not allow the Patient ID of an existing patient to be changed and shall reject an update request for a Patient ID that does not exist.	Functional	Medium	Hospital Staff; UC-03; BR-01	The update menu offers no Patient ID edit; updating a non-existent ID displays “Patient not found” and changes nothing. Test: TC-PAT-07	Depends on HMS-F-006

4.2 Doctor Management - Functional Requirements

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-008	The system shall allow Hospital Staff to add a doctor by entering Doctor ID, name, specialization, department, contact number and availability information.	Functional	High	Hospital Staff; UC-04; BR-02	Valid data creates exactly one new doctor record, a confirmation is displayed, and doctors.csv contains the record after the operation. Test: TC-DOC-01	HospitalSystem, Doctor, FileManager
HMS-F-009	The system shall reject addition of a doctor whose Doctor ID already exists and shall leave all stored records unchanged.	Functional	High	Hospital Staff; UC-04; BR-02	Adding an existing ID displays “Doctor ID already exists”; doctor count and file content are identical before and after. Test: TC-DOC-02	Depends on HMS-F-008
HMS-F-010	The system shall validate every doctor field against rules VR-01, VR-02, VR-05, VR-08, VR-09 and VR-14 before storing it, and shall reject an invalid value with a message naming the field and the rule violated.	Functional	High	Hospital Staff; UC-04; BR-08	Boundary values for each field are accepted or rejected as defined in Table 4-1; no record is stored while any field is invalid. Test: TC-DOC-03	Table 4-1
HMS-F-011	The system shall allow Hospital Staff to search doctors by Doctor ID (exact match), by name (case-insensitive partial match) or by specialization (case-insensitive partial match) and shall display all matching doctors.	Functional	High	Hospital Staff; UC-05	Each of the three criteria returns exactly the matching doctors; no match displays “No matching doctor found”. Test: TC-DOC-04	—
HMS-F-012	The system shall display all stored fields of a selected doctor and shall provide a list of all doctors showing Doctor ID, name, specialization and department.	Functional	High	Hospital Staff; UC-05	Selected doctor view shows all six fields as stored; list view shows every stored doctor once; an empty dataset displays “No doctors registered”. Test: TC-DOC-05	Depends on HMS-F-008
HMS-F-013	The system shall allow Hospital Staff to update the name, specialization, department, contact number and availability information of an existing doctor, applying the same validation rules as addition.	Functional	Medium	Hospital Staff; UC-06; BR-08	A valid change is reflected on the next view and in doctors.csv; an invalid value is rejected and the stored record is unchanged. Test: TC-DOC-06	Depends on HMS-F-008, HMS-F-010
HMS-F-014	The system shall not allow the Doctor ID of an existing doctor to be changed and shall reject an update request for a Doctor ID that does not exist.	Functional	Medium	Hospital Staff; UC-06; BR-02	The update menu offers no Doctor ID edit; updating a non-existent ID displays “Doctor not found” and changes nothing. Test: TC-DOC-07	Depends on HMS-F-013

4.3 Appointment Management - Booking Requirements

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-015	The system shall allow Hospital Staff to book an appointment by entering Patient ID, Doctor ID, appointment date and appointment time, and shall create a record with a system-generated unique Appointment ID and status Booked.	Functional	High	Hospital Staff; UC-07; BR-03, BR-04, BR-05	A valid booking displays the new Appointment ID, appears in the appointment list with status Booked, and is present in appointments.csv. Test: TC-APT-01	Depends on HMS-F-001, HMS-F-008
HMS-F-016	The system shall reject a booking whose Patient ID does not match a registered patient and shall create no appointment record.	Functional	High	Hospital Staff; UC-07; BR-03	Booking with an unregistered Patient ID displays “Patient not found”; appointment count and file content are unchanged. Test: TC-APT-02	Depends on HMS-F-015
HMS-F-017	The system shall reject a booking whose Doctor ID does not match a registered doctor and shall create no appointment record.	Functional	High	Hospital Staff; UC-07; BR-04	Booking with an unregistered Doctor ID displays “Doctor not found”; appointment count and file content are unchanged. Test: TC-APT-03	Depends on HMS-F-015
HMS-F-018	The system shall validate the appointment date and time against rules VR-10 and VR-11 and shall reject an invalid value with an error message.	Functional	High	Hospital Staff; UC-07; BR-08	Inputs such as 2026-02-30, 2026-13-01, 24:00 and 9:5 are rejected; 2026-10-05 and 09:30 are accepted. Test: TC-APT-04	Table 4-1

4.3 Appointment Management - Availability, View and Cancellation

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-019	The system shall reject a booking when the selected doctor already has a non-cancelled appointment at the same date and time, and shall create no appointment record.	Functional	High	Hospital Staff; UC-07, UC-10; BR-05	A second booking for the same Doctor ID, date and time displays “Doctor not available at the requested date and time”; appointment count is unchanged. Test: TC-APT-05, TC-NFR-08	Depends on HMS-F-015
HMS-F-020	The system shall allow Hospital Staff to check doctor availability using Doctor ID, date and time, report Available/Not Available, and not modify records.	Functional	High	Hospital Staff; UC-10; BR-05	Valid doctor/date/time returns the stored-slot result; invalid doctor/date/time is rejected; file unchanged. Test: TC-APT-06	HMS-F-008, HMS-F-018
HMS-F-021	The system shall exclude appointments with status Cancelled when determining doctor availability.	Functional	High	Hospital Staff; UC-07, UC-09, UC-10; BR-05, BR-06	After an appointment is cancelled, a new booking for the same doctor, date and time is accepted. Test: TC-APT-07	Depends on HMS-F-023
HMS-F-022	The system shall allow Hospital Staff to view all appointments, or only the appointments of a given Patient ID or Doctor ID, sorted by date then time in ascending order, showing Appointment ID, Patient ID and name, Doctor ID and name, date, time and status.	Functional	High	Hospital Staff; UC-08	Each of the three views lists exactly the expected records in date/time order and includes Cancelled records with status Cancelled; no records displays “No appointments found”. Test: TC-APT-08	Depends on HMS-F-015
HMS-F-023	The system shall allow Hospital Staff to cancel an appointment by Appointment ID and, after staff confirmation, shall change its status to Cancelled while retaining the record.	Functional	Medium	Hospital Staff; UC-09; BR-06	After cancellation the record is still listed with status Cancelled and appointments.csv still contains the record; declining the confirmation changes nothing. Test: TC-APT-09	Depends on HMS-F-015
HMS-F-024	The system shall reject cancellation of an Appointment ID that does not exist or that already has status Cancelled, and shall leave all records unchanged.	Functional	Medium	Hospital Staff; UC-09; BR-06	Cancelling a non-existent ID displays “Appointment not found”; cancelling an already cancelled ID displays “Appointment already cancelled”; no record changes. Test: TC-APT-10	Depends on HMS-F-023

4.4 Persistence and File Handling - Loading

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-025	The system shall load the records of patients.csv, doctors.csv and appointments.csv from the data directory into memory at startup.	Functional	High	Hospital Staff; All UCs; BR-09	With valid files containing N records each, the system lists N patients, N doctors and N appointments immediately after startup. Test: TC-PER-01	FileManager
HMS-F-026	The system shall treat a missing data file as an empty dataset, shall continue to run, and shall create the file on the next successful save.	Functional	High	Hospital Staff; All UCs; BR-09	Startup with any or all data files absent shows the main menu with no error termination; the file exists after the first successful add or booking. Test: TC-PER-02	FileManager
HMS-F-027	The system shall skip each malformed record (wrong number of fields or a field violating Table 4-1) during loading, shall continue loading the remaining records, and shall display the number of records skipped per file.	Functional	High	Hospital Staff; All UCs; BR-10	A file with 10 valid and 3 malformed lines loads 10 records, displays “3 malformed record(s) skipped”, and does not terminate. Test: TC-PER-03	Depends on HMS-F-025
HMS-F-028	The system shall skip, during loading, any record whose ID duplicates an already loaded record, and any appointment record whose Patient ID or Doctor ID does not exist among the loaded records.	Functional	Medium	Hospital Staff; All UCs; BR-01, BR-02, BR-03, BR-04	Duplicate-ID lines and orphan appointment lines are not loaded and are included in the skipped-record count. Test: TC-PER-04	Depends on HMS-F-027

4.4 Persistence and File Handling - Saving / Recovery

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-029	The system shall write the affected data file after each successful registration, addition, update, booking or cancellation and before returning to the menu.	Functional	High	Hospital Staff; UC-01, UC-03, UC-04, UC-06, UC-07, UC-09	Immediately after each modifying operation, the corresponding CSV file contains the change. Test: TC-PER-05	FileManager
HMS-F-030	The system shall make all patient, doctor and appointment records available after the application is closed and restarted.	Functional	High	Hospital Staff; All UCs	Following the demonstration workflow (register, add, book, exit, restart), all records and the appointment status are displayed unchanged. Test: TC-PER-06	Depends on HMS-F-025, HMS-F-029
HMS-F-031	If a data file cannot be written, the system shall report the failure, remain running, and preserve the previously valid stored state.	Functional	Medium	Hospital Staff; All UCs	With a read-only target, failure is reported, previous file content remains valid, and the menu remains usable. Test: TC-PER-07	Depends on HMS-F-029

4.5 Console User Interface and Input Handling

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance / Test	Dependencies
HMS-F-032	The system shall display a main menu with the options 1 Patient Management, 2 Doctor Management, 3 Appointment Management and 4 Exit, and shall display a submenu with a return option for options 1 to 3.	Functional	High	Hospital Staff; All UCs	Selecting 1, 2 or 3 displays the submenu of that module listing the operations of Section 3.1; each submenu returns to the main menu. Test: TC-UI-01	Console UI
HMS-F-033	The system shall reject a menu selection that is empty, non-numeric or outside the listed range, shall display an error message, and shall re-display the same menu.	Functional	High	Hospital Staff; All UCs; BR-08	Inputs “9”, “abc”, “-1” and an empty line at each menu produce an error message and the same menu; the application does not terminate. Test: TC-UI-02	Console UI
HMS-F-034	The system shall not modify any in-memory or stored record when any input of the operation fails validation.	Functional	High	Hospital Staff; All UCs; BR-08	For each modifying operation, entering an invalid value leaves the affected CSV file byte-identical and the in-memory records unchanged. Test: TC-UI-03	HospitalSystem
HMS-F-035	The system shall end every operation with either a success confirmation or an error message stating the reason for failure.	Functional	Medium	Hospital Staff; All UCs	Every operation in the functional test suite displays exactly one of the two outcomes. Test: TC-UI-04	Console UI
HMS-F-036	The system shall terminate normally when option 4 Exit is selected, with all previously confirmed changes already stored.	Functional	Medium	Hospital Staff; All UCs	After Exit, the process ends with exit code 0 and the data files contain all confirmed changes. Test: TC-UI-05	Console UI

5. Non-functional requirements

Req ID	Requirement	Category	Priority	Acceptance / Measurement
HMS-NF-001	The system shall not terminate abnormally for any invalid keyboard input; for each of at least 20 defined invalid inputs (empty, non-numeric, over-length, special characters) it shall display an error message and return to a prompt.	Reliability	High	0 crashes across the 20 invalid-input test cases. Test: TC-NFR-01
HMS-NF-002	The system shall start successfully for any combination of missing, empty or malformed data files and shall display the main menu.	Reliability	High	Startup succeeds in all tested combinations (all missing; all empty; 5 malformed lines per file). Test: TC-NFR-02
HMS-NF-003	The system shall retain 100% of records confirmed as saved after a normal exit and restart.	Persistence	High	Record counts and field values before exit equal those after restart in the demonstration workflow. Test: TC-NFR-03
HMS-NF-004	With 1000 patient, 1000 doctor and 1000 appointment records, the system shall complete startup loading within 3 seconds and each search, view, booking (including conflict check) and cancellation within 2 seconds, excluding user typing time.	Performance	Low	Measured times on a reference PC (4 GB RAM, dual-core CPU) are within the limits in 3 of 3 runs. Test: TC-NFR-04
HMS-NF-005	Every prompt shall state the expected input format, every error message shall name the field and the rule violated, and any operation shall be reachable within at most 3 menu selections from the main menu.	Usability	High	Checklist review of all prompts and messages passes 100%; menu depth of every operation ≤ 3. Test: TC-NFR-05
HMS-NF-006	The source code shall separate responsibilities into the classes Patient, Doctor, Appointment, HospitalSystem and FileManager, and only FileManager shall perform file input/output.	Maintainability	Medium	Code review confirms the five classes; a search of the source finds file-stream usage only in FileManager. Test: TC-NFR-06
HMS-NF-007	The system shall compile without errors using g++ (GCC 9 or later / MinGW-w64) with the option -std=c++17 and shall run on Windows 10 or later and on Linux.	Portability	Low	Clean build and execution of the functional smoke test on both platforms. Test: TC-NFR-07
HMS-NF-008	The system shall detect doctor appointment conflicts with 100% accuracy for the defined conflict test matrix (same doctor/same slot, same doctor/different time, different doctor/same slot, same doctor/same slot after cancellation).	Correctness	High	All four matrix cases give the expected accept/reject result. Test: TC-NFR-08
HMS-NF-009	The system shall run as a single-user standalone process on one machine, without a network connection, database server or third-party library beyond the C++ standard library.	Deployment / Compatibility	Medium	The application runs on a machine with networking disabled and with only the compiler runtime installed. Test: TC-NFR-09
HMS-NF-010	Each data file shall be a plain-text, delimiter-separated file with a header row and one record per line, readable in a text editor.	Maintainability / Interoperability	Low	Each file opens in a text editor and matches the layouts of Section 3.3. Test: TC-NFR-10

5.1 Security objectives

Objective	Name	Description	Realised by
HMS-SO-01	Integrity	Validated operations respect business rules; malformed input/write failures shall not corrupt records.	SR-001, 002, 005, 006, 010
HMS-SO-02	Availability / Resilience	Application remains operational on invalid input and missing/malformed files.	SR-003, 004, 005

Objective	Name	Description	Realised by
HMS-SO-03	Confidentiality	Limit patient-data exposure to the local machine and minimum needed display.	SR-007, 008, 009, 010

5.1.2 Security requirements

Req ID	Requirement (shall...)	Priority	Objective	Acceptance / Test
HMS-SR-001	The system shall validate all keyboard input for type, length, format and range against Table 4-1 using an accept-list approach before the input is used.	High	HMS-SO-01	Boundary values accepted/rejected as defined. TC-SEC-01
HMS-SR-002	The system shall reject text input that contains a comma, carriage-return or line-feed character, or that begins with =, +, - or @.	High	HMS-SO-01	Delimiter/formula inputs rejected; no record stored. TC-SEC-02
HMS-SR-003	The system shall accept input lines of any length without memory corruption and shall reject any value longer than the maximum length in Table 4-1.	High	HMS-SO-02	10,000-character input rejected; application continues. TC-SEC-03
HMS-SR-004	The system shall treat data files as untrusted input, shall validate every field of every loaded record with the rules of Table 4-1, and shall skip an invalid record entirely rather than load it partially.	High	HMS-SO-02	Malformed file rows skipped; only valid records load. TC-SEC-04
HMS-SR-005	The system shall not discard the previous content of a data file until the complete new content has been written successfully, and shall keep the previous content and report an error if the write fails.	High	HMS-SO-01, HMS-SO-02	Simulated write failure preserves previous file. TC-SEC-05
HMS-SR-006	The system shall not physically delete an appointment record by any mandatory function; cancellation shall retain the record with status Cancelled.	High	HMS-SO-01	Cancellation retains record count/status Cancelled. TC-SEC-06
HMS-SR-007	The system shall show only identifiers and names of the patient and the doctor in appointment views and shall not show the patient's contact number, address or medical history note there.	Medium	HMS-SO-03	Appointment views exclude contact/address/medical note. TC-SEC-07
HMS-SR-008	The system shall not open network connections or transmit patient, doctor or appointment data outside the local machine.	Medium	HMS-SO-03	Code/test run shows no network transmission. TC-SEC-08
HMS-SR-009	The system shall not display internal information such as file-system paths, raw record contents or exception text in any error message.	Medium	HMS-SO-03	Failure messages expose no path/raw-record/exception detail. TC-SEC-09
HMS-SR-010	The system shall read and write data files only in its designated data directory and shall not accept a file name or path from the user.	Medium	HMS-SO-01, HMS-SO-03	No path input; files only in designated data directory. TC-SEC-10
HMS-SR-011	If Administrator Login is implemented, password shall not be stored in plain text and login shall be blocked after 3 consecutive failed attempts.	Optional	HMS-SO-01	Optional: no plain-text password; 4th attempt refused after 3 failures. TC-OPT-04

6. Quality attributes & acceptance criteria

Attribute	Requirements	Measure
Reliability	NF-001, NF-002, F-033, F-034	0 crashes on invalid-input and bad-file tests
Persistence	NF-003, F-025 to F-031	100% saved records retrievable after restart
Usability	NF-005, F-032, F-035	Prompt/message checklist 100%; menu depth <= 3
Performance	NF-004	<= 2 s operations; <= 3 s startup at 1000 records
Correctness	NF-008, F-019, F-021	100% conflict test matrix
Maintainability	NF-006, NF-010	Five-class separation; file I/O only in FileManager
Portability	NF-007, NF-009	C++17 build/run on Windows and Linux; standalone
Security	SR-001 to SR-010	All mandatory security test cases pass

Exit criteria: all High/Medium mandatory functional requirements verified; no High-priority NFR/security failure; end-to-end baseline demo passes; mandatory RTM tests pass. Optional features are excluded.

7. System models and diagrams

Actor/system boundary: Hospital Staff / Administrator is the only actor. Patient and Doctor are internal records, and the CSV files are internal persistence. The two UML use-case diagrams below are retained from the baseline SRS.

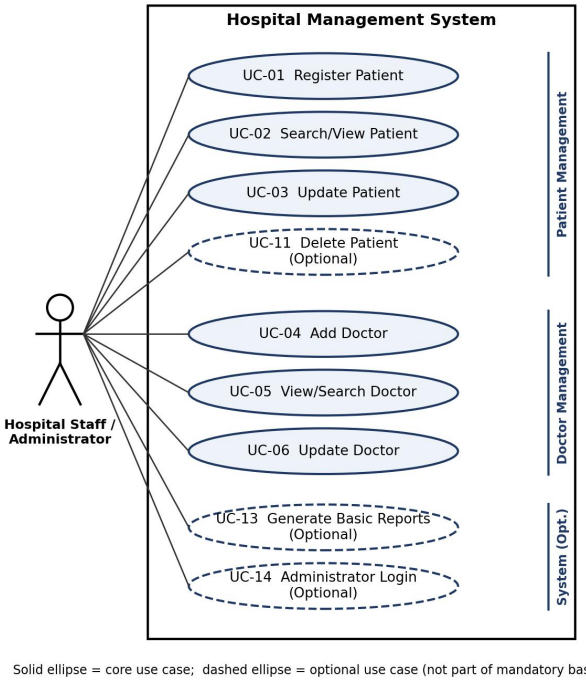


Figure 1: Patient/Doctor use cases

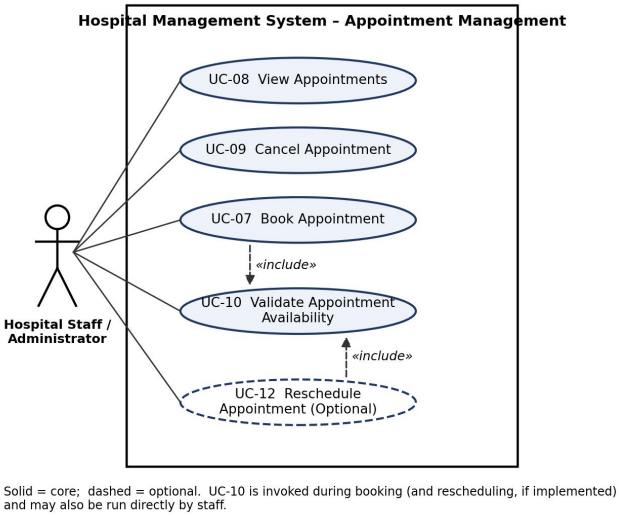


Figure 2: Appointment use cases

7.3 Use case summary

UC	Use case	Priority	Module	Requirements
UC-01	Register Patient	High	Patient Management	HMS-F-001 to 003, 029
UC-02	Search/View Patient	High	Patient Management	HMS-F-004, 005
UC-03	Update Patient	Medium	Patient Management	HMS-F-006, 007
UC-04	Add Doctor	High	Doctor Management	HMS-F-008 to 010, 029
UC-05	View/Search Doctor	High	Doctor Management	HMS-F-011, 012
UC-06	Update Doctor	Medium	Doctor Management	HMS-F-013, 014
UC-07	Book Appointment	High	Appointment Management	HMS-F-015 to 019, 021, 029
UC-08	View Appointments	High	Appointment Management	HMS-F-022, HMS-SR-007
UC-09	Cancel Appointment	Medium	Appointment Management	HMS-F-021, 023, 024, HMS-SR-006
UC-10	Validate Appointment Availability	High	Appointment Management	HMS-F-019 to 021, HMS-NF-008
UC-11	Delete Patient	Optional	Patient Management	HMS-O-001
UC-12	Reschedule Appointment	Optional	Appointment Management	HMS-O-002
UC-13	Generate Basic Reports	Optional	System-level	HMS-O-003
UC-14	Administrator Login	Optional	System-level	HMS-O-004, HMS-SR-011

7.4 Use case specifications - UC-01 to UC-05

UC	Name	Preconditions	Main flow	Alternate / exception	Postconditions	Related reqs
UC-01	Register Patient	Application is running and the Patient Management submenu is displayed.	1. Staff selects Register Patient. 2. System prompts for Patient ID, name, age, gender, contact number, address and optional medical history note. 3. Staff enters the values. 4. System validates each value against Table 4-1. 5. System verifies that the Patient ID is not already registered. 6. System stores the record and writes patients.csv. 7. System displays a confirmation.	4a. A value is invalid: system displays the error and re-prompts for that field. 5a. Patient ID exists: system rejects the registration and changes nothing. 6a. File cannot be written: system displays the error of HMS-F-031.	Success: the patient record exists in memory and in patients.csv. Failure: no record is changed.	HMS-F-001 to 003, 029
UC-02	Search/View Patient	Patient Management submenu is displayed.	1. Staff selects Search Patient (or View All Patients). 2. System asks for the search criterion (Patient ID or name) and value. 3. Staff enters the value. 4. System displays the matching patients. 5. Staff optionally selects a patient; system displays all stored fields.	4a. No match: system displays “No matching patient found”. 2a. Empty or invalid search value: system rejects it and re-prompts.	No record is changed.	HMS-F-004, 005
UC-03	Update Patient	Patient Management submenu is displayed.	1. Staff selects Update Patient and enters the Patient ID. 2. System displays the current record. 3. Staff selects a field and enters a new value. 4. System validates the value. 5. System updates the record, writes patients.csv and confirms.	1a. Patient ID not found: system displays “Patient not found”. 4a. Value invalid: system rejects it; the record is unchanged.	Success: updated record stored. Failure: record unchanged. Patient ID never changes.	HMS-F-006, 007
UC-04	Add Doctor	Doctor Management submenu is displayed.	1. Staff selects Add Doctor. 2. System prompts for Doctor ID, name, specialization, department, contact number and availability information. 3. Staff enters the values. 4. System validates each value against Table 4-1. 5. System verifies that the Doctor ID is not already registered. 6. System stores the record and writes doctors.csv. 7. System displays a confirmation.	4a. A value is invalid: system displays the error and re-prompts for that field. 5a. Doctor ID exists: system rejects the addition and changes nothing. 6a. File cannot be written: system displays the error of HMS-F-031.	Success: the doctor record exists in memory and in doctors.csv. Failure: no record is changed.	HMS-F-008 to 010, 029
UC-05	View/Search Doctor	Doctor Management submenu is displayed.	1. Staff selects Search Doctor (or View All Doctors). 2. System asks for the criterion (Doctor ID, name or specialization) and value. 3. Staff enters the value. 4. System displays the matching doctors. 5. Staff optionally selects a doctor; system displays all stored fields.	4a. No match: system displays “No matching doctor found”. 2a. Empty or invalid search value: system rejects it and re-prompts.	No record is changed.	HMS-F-011, 012

7.4 Use case specifications - UC-06 to UC-10

UC	Name	Preconditions	Main flow	Alternate / exception	Postconditions	Related reqs
UC-06	Update Doctor	Doctor Management submenu is displayed.	1. Staff selects Update Doctor and enters the Doctor ID. 2. System displays the current record. 3. Staff selects a field and enters a new value. 4. System validates the value. 5. System updates the record, writes doctors.csv and confirms.	1a. Doctor ID not found: system displays “Doctor not found”. 4a. Value invalid: system rejects it; the record is unchanged.	Success: updated record stored. Failure: record unchanged. Doctor ID never changes.	HMS-F-013, 014
UC-07	Book Appointment	Appointment Management submenu is displayed; the patient and the doctor are registered.	1. Staff selects Book Appointment. 2. System prompts for Patient ID, Doctor ID, date and time. 3. Staff enters the values. 4. System verifies that the patient exists and that the doctor exists. 5. System validates date and time formats. 6. System performs UC-10 Validate Appointment Availability for the doctor, date and time («include»). 7. System creates the appointment with a new Appointment ID and status Booked, writes appointments.csv and displays the Appointment ID.	4a. Patient or doctor not found: system rejects the booking. 5a. Date or time invalid: system rejects the value and re-prompts. 6a. Doctor not available: system rejects the booking and staff may choose another time. 7a. File cannot be written: system displays the error of HMS-F-031.	Success: one new Booked appointment stored. Failure: no record is changed.	HMS-F-015 to 019, 021, 029
UC-08	View Appointments	Appointment Management submenu is displayed.	1. Staff selects View Appointments. 2. System asks for the view: all, by Patient ID or by Doctor ID. 3. Staff selects the view and, if required, enters the ID. 4. System displays the matching appointments sorted by date then time, including Cancelled ones.	4a. No appointments match: system displays “No appointments found”. 3a. ID not registered: system displays the corresponding not-found message.	No record is changed.	HMS-F-022, HMS-SR-007
UC-09	Cancel Appointment	Appointment Management submenu is displayed.	1. Staff selects Cancel Appointment and enters the Appointment ID. 2. System displays the appointment summary and asks for confirmation. 3. Staff confirms. 4. System sets the status to Cancelled, writes appointments.csv and confirms.	1a. Appointment ID not found: system displays “Appointment not found”. 1b. Appointment already Cancelled: system displays “Appointment already cancelled”. 3a. Staff declines: system changes nothing.	Success: status is Cancelled and the record is retained. Failure or decline: no record is changed.	HMS-F-021, 023, 024, HMS-SR-006
UC-10	Validate Appointment Availability	Doctor is registered. Invoked from UC-07 (and UC-12 if implemented) or run directly from the Appointment Management submenu.	1. System obtains Doctor ID, date and time (entered by staff for a direct check, or passed on by the calling use case). 2. System verifies that the doctor exists and that date and time are valid. 3. System searches the appointments of that doctor for a non-cancelled appointment at the same date and time. 4. System returns Available (none found) or Not Available (one found).	2a. Doctor not found or date/time invalid: system displays the error; no result is returned.	No record is changed.	HMS-F-019 to 021, HMS-NF-008

Optional use cases - concise specifications

UC	Name	Precondition	Flow / rule	Requirements
UC-11	Delete Patient	Patient exists; optional feature enabled.	Delete only after confirmation and only if no appointment record references the patient.	HMS-O-001; BR-07
UC-12	Reschedule Appointment	Booked appointment exists; optional feature enabled.	Validate new date/time and reuse booking conflict rules; keep Appointment ID.	HMS-O-002; BR-11
UC-13	Generate Basic Reports	Optional reporting enabled.	List a selected doctor’s appointments for a selected date and show Booked/Cancelled counts.	HMS-O-003
UC-14	Administrator Login	Optional login enabled.	Require credentials before main menu; wrong credentials denied; apply SR-011.	HMS-O-004; HMS-SR-011

8. Requirements Traceability Matrix (RTM) - Functional Part 1

Req ID	Requirement (short)	Section / Design	Module	Test case(s)	Status	Comments
HMS-F-001	Register patient	4.1 / DS-PAT-01	Patient Management	TC-PAT-01	N	UC-01; BR-01
HMS-F-002	Reject duplicate patient ID	4.1 / DS-PAT-02	Patient Management	TC-PAT-02	N	UC-01; BR-01
HMS-F-003	Validate patient fields	4.1 / DS-PAT-03	Patient Management	TC-PAT-03	N	UC-01; BR-08
HMS-F-004	Search patient	4.1 / DS-PAT-04	Patient Management	TC-PAT-04	N	UC-02
HMS-F-005	View patient / list patients	4.1 / DS-PAT-05	Patient Management	TC-PAT-05	N	UC-02
HMS-F-006	Update patient	4.1 / DS-PAT-06	Patient Management	TC-PAT-06	N	UC-03; BR-08
HMS-F-007	Patient ID immutable / not found	4.1 / DS-PAT-07	Patient Management	TC-PAT-07	N	UC-03; BR-01
HMS-F-008	Add doctor	4.2 / DS-DOC-01	Doctor Management	TC-DOC-01	N	UC-04; BR-02
HMS-F-009	Reject duplicate doctor ID	4.2 / DS-DOC-02	Doctor Management	TC-DOC-02	N	UC-04; BR-02
HMS-F-010	Validate doctor fields	4.2 / DS-DOC-03	Doctor Management	TC-DOC-03	N	UC-04; BR-08
HMS-F-011	Search doctor	4.2 / DS-DOC-04	Doctor Management	TC-DOC-04	N	UC-05
HMS-F-012	View doctor / list doctors	4.2 / DS-DOC-05	Doctor Management	TC-DOC-05	N	UC-05
HMS-F-013	Update doctor	4.2 / DS-DOC-06	Doctor Management	TC-DOC-06	N	UC-06; BR-08
HMS-F-014	Doctor ID immutable / not found	4.2 / DS-DOC-07	Doctor Management	TC-DOC-07	N	UC-06; BR-02
HMS-F-015	Book appointment	4.3 / DS-APT-01	Appointment Management	TC-APT-01	N	UC-07; BR-03, BR-04, BR-05
HMS-F-016	Booking requires existing patient	4.3 / DS-APT-02	Appointment Management	TC-APT-02	N	UC-07; BR-03
HMS-F-017	Booking requires existing doctor	4.3 / DS-APT-03	Appointment Management	TC-APT-03	N	UC-07; BR-04
HMS-F-018	Validate date and time	4.3 / DS-APT-04	Appointment Management	TC-APT-04	N	UC-07; BR-08

Status legend: N = not yet verified, P = passed testing, A = accepted. “Final” describes the requirements baseline; RTM status records verification progress.

8. Requirements Traceability Matrix (RTM) - Functional Part 2

Req ID	Requirement (short)	Section / Design	Module	Test case(s)	Status	Comments
HMS-F-019	Reject conflicting booking	4.3 / DS-APT-05	Appointment Management	TC-APT-05, TC-NFR-08	N	UC-07, UC-10; BR-05
HMS-F-020	Check availability	4.3 / DS-APT-06	Appointment Management	TC-APT-06	N	UC-10; BR-05
HMS-F-021	Cancelled slot is re-bookable	4.3 / DS-APT-07	Appointment Management	TC-APT-07	N	UC-07, UC-09, UC-10; BR-05, BR-06
HMS-F-022	View appointments	4.3 / DS-APT-08	Appointment Management	TC-APT-08	N	UC-08
HMS-F-023	Cancel appointment	4.3 / DS-APT-09	Appointment Management	TC-APT-09	N	UC-09; BR-06
HMS-F-024	Reject invalid cancellation	4.3 / DS-APT-10	Appointment Management	TC-APT-10	N	UC-09; BR-06
HMS-F-025	Load records at startup	4.4 / DS-PER-01	Persistence (FileManager)	TC-PER-01	N	All; BR-09
HMS-F-026	Missing file treated as empty	4.4 / DS-PER-02	Persistence (FileManager)	TC-PER-02	N	All; BR-09
HMS-F-027	Skip malformed records	4.4 / DS-PER-03	Persistence (FileManager)	TC-PER-03	N	All; BR-10
HMS-F-028	Skip duplicate / orphan records	4.4 / DS-PER-04	Persistence (FileManager)	TC-PER-04	N	All; BR-01, BR-02, BR-03, BR-04
HMS-F-029	Save after each change	4.4 / DS-PER-05	Persistence (FileManager)	TC-PER-05	N	UC-01, UC-03, UC-04, UC-06, UC-07, UC-09
HMS-F-030	Records persist across restarts	4.4 / DS-PER-06	Persistence (FileManager)	TC-PER-06	N	All
HMS-F-031	Report write failure	4.4 / DS-PER-07	Persistence (FileManager)	TC-PER-07	N	All
HMS-F-032	Main menu and submenus	4.5 / DS-UI-01	Console UI	TC-UI-01	N	All
HMS-F-033	Validate menu selection	4.5 / DS-UI-02	Console UI	TC-UI-02	N	All; BR-08
HMS-F-034	Invalid input never modifies records	4.5 / DS-UI-03	Console UI	TC-UI-03	N	All; BR-08
HMS-F-035	Outcome message for every operation	4.5 / DS-UI-04	Console UI	TC-UI-04	N	All
HMS-F-036	Normal exit	4.5 / DS-UI-05	Console UI	TC-UI-05	N	All

8. Requirements Traceability Matrix - NFR, Security and Optional Requirements

Req ID	Requirement (short)	Section / Design	Module	Test	Status
HMS-NF-001	No crash on invalid input	5 / DS-NFR-01	All modules	TC-NFR-01	N
HMS-NF-002	Robust startup	5 / DS-NFR-02	Persistence (FileManager)	TC-NFR-02	N
HMS-NF-003	Data durability	5 / DS-NFR-03	Persistence (FileManager)	TC-NFR-03	N
HMS-NF-004	Responsive at 1000 records	5 / DS-NFR-04	All modules	TC-NFR-04	N
HMS-NF-005	Clear prompts and messages	5 / DS-NFR-05	Console UI	TC-NFR-05	N
HMS-NF-006	Modular class structure	5 / DS-NFR-06	All modules	TC-NFR-06	N
HMS-NF-007	GCC/MinGW portability	5 / DS-NFR-07	All modules	TC-NFR-07	N
HMS-NF-008	Conflict detection accuracy	5 / DS-NFR-08	Appointment Management	TC-NFR-08	N
HMS-NF-009	Standalone operation	5 / DS-NFR-09	All modules	TC-NFR-09	N
HMS-NF-010	Human-readable data files	5 / DS-NFR-10	Persistence (FileManager)	TC-NFR-10	N
HMS-SR-001	Accept-list input validation	5.1.2 / DS-SEC-01	All modules	TC-SEC-01	N
HMS-SR-002	Reject delimiter / formula characters	5.1.2 / DS-SEC-02	All modules	TC-SEC-02	N
HMS-SR-003	Bounded input handling	5.1.2 / DS-SEC-03	Console UI	TC-SEC-03	N
HMS-SR-004	Untrusted data files	5.1.2 / DS-SEC-04	Persistence (FileManager)	TC-SEC-04	N
HMS-SR-005	Safe file write	5.1.2 / DS-SEC-05	Persistence (FileManager)	TC-SEC-05	N
HMS-SR-006	Appointment records retained	5.1.2 / DS-SEC-06	Appointment Management	TC-SEC-06	N
HMS-SR-007	Minimal data in appointment views	5.1.2 / DS-SEC-07	Appointment Management	TC-SEC-07	N
HMS-SR-008	No network transmission	5.1.2 / DS-SEC-08	All modules	TC-SEC-08	N
HMS-SR-009	No internal detail in errors	5.1.2 / DS-SEC-09	All modules	TC-SEC-09	N
HMS-SR-010	Fixed data directory	5.1.2 / DS-SEC-10	Persistence (FileManager)	TC-SEC-10	N
HMS-SR-011	Login credential protection	5.1.2 / DS-SEC-11	Console UI	TC-OPT-04	N
HMS-O-001	Delete patient	4.6 / DS-OPT-01	Patient Management	TC-OPT-01	N
HMS-O-002	Reschedule appointment	4.6 / DS-OPT-02	Appointment Management	TC-OPT-02	N
HMS-O-003	Basic reports	4.6 / DS-OPT-03	As per feature	TC-OPT-03	N
HMS-O-004	Administrator login	4.6 / DS-OPT-04	As per feature	TC-OPT-04	N

8.2 Coverage and count summary

Category	Count	Notes
Functional requirements (HMS-F)	36	High 26; Medium 10
Non-functional requirements (HMS-NF)	10	Measurable NFRs
Security objectives (HMS-SO)	3	Integrity, Availability/Resilience, Confidentiality
Security requirements (HMS-SR)	10 mandatory + 1 optional	SR-011 applies only to optional UC-14
Optional features (HMS-O)	4	Excluded from mandatory exit criteria
Use cases	14	UC-01 to UC-10 mandatory; UC-11 to UC-14 optional
Mandatory acceptance tests	56	PAT 7; DOC 7; APT 10; PER 7; UI 5; NFR 10; SEC 10